
PROJECT REPORT: CIS3186- Mobile Device Programming

Project n°[1]: SOMMLY

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GitHub Link: https://github.com/PierreMayet/Sommly_CIS3186_MobileProg

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I. Introduction

With mobile commerce sales projected to account for over 53% of all global e-commerce by 2025, the digital transformation of the wine industry is no longer an option but a necessity. Despite this growth, recent consumer surveys indicate that most wine buyers feel overwhelmed when trying to choose a bottle, often struggling to understand complex labels or find the right food pairing.

In the context of our mobile programming project supervised by Dr. Conrad Attard, we have developed Sommyly. This application allows merging a high-end shopping experience with smart technology.

To successfully carry out our project, we are subject to a specific set of requirements. First, the primary objective is to develop a mobile commerce application that addresses the needs of a specific domain. To achieve this, we are required to understand and analyze the stakeholder's requirements to ensure the solution is truly user centric. In addition, we must propose a functional first prototype available through a mobile device, following best practices in interface design. Furthermore, the core of the project is to design an app-based IoT solution for this domain, enabling real-time interaction between the user and connected objects. Finally, the project requires us to evaluate our designs through a formal usability study, collect feedback, and performance metrics to improve the final application.

II. Problem statement and objectives

2.1 - Problem Statement

The wine market presents a unique challenge for consumers due to its diversity, complexity, and the subjective nature of taste. In both physical retail environments and online platforms, wine consumers are often confronted with an overwhelming number of options. This abundance, while beneficial in theory, frequently leads to decision fatigue and uncertainty, particularly for users who lack extensive wine knowledge.

Novice consumers, in particular, struggle to select wines that align with their personal preferences or complement specific meals. Traditional sources of guidance, such as sommeliers or in-store advisors, are not always available, and existing digital solutions often rely on complex terminology or require prior expertise. As a result, many users feel excluded from the wine discovery process and may resort to repetitive or uninformed purchasing decisions.

On the other hand, experienced wine enthusiasts face different limitations. While they may possess sufficient knowledge to evaluate wines, they often lack efficient tools that facilitate the discovery of niche products or independent producers. Mainstream platforms tend to promote popular or mass-market wines, reducing visibility for smaller producers, and limiting meaningful exploration. Beyond the purchasing phase, another critical issue lies in post-purchase wine preservation. Wine quality is highly sensitive to environmental conditions such as temperature and humidity. Most consumers lack the means to monitor these factors, which can negatively impact wine integrity over time. The absence of accessible monitoring solutions creates a gap between purchase and consumption, potentially diminishing the overall user experience.

These challenges highlight the need for a mobile solution that not only simplifies wine selection but also supports informed decision-making, discovery, and quality assurance throughout the entire user journey.

2.2 - Objectives

The Sommly app was designed to address the issues identified by offering a user-centric solution based on modern technologies. Its main objective is to simplify wine selection through personalised recommendations tailored to users' preferences and consumption context, particularly food and wine pairings. By reducing the complexity of the selection process, the application aims to improve user confidence and make the shopping experience more enjoyable.

Another important objective of the project is to offer effective filtering tools to facilitate navigation within the wine catalogue. Filters based on wine type, food pairings and personal preferences allow users to quickly find products that meet their expectations, whether they are novices or more experienced.

Sommly also seeks to strengthen the link between consumers and wine producers. By associating each wine with its producer and offering a more direct connection, the application promotes transparency and allows users to access more detailed information about the origin of the products. This approach also helps to promote producers, particularly independent or small-scale structures.

In addition, the project incorporates an Internet of Things (IoT) dimension to address the challenges of wine storage after purchase. Real-time monitoring of parameters such as temperature and humidity allows users to better preserve the quality of their bottles. Secure access to this data ensures personalised use while respecting the confidentiality of the information.

Finally, the project pays particular attention to best practices in UX/UI design for mobile applications. Sommly is based on recognised recommendations, such as Material Design principles and mobile accessibility standards, in order to offer an application that is easy to use, consistent and accessible to as many people as possible.

III.App Overview

Sommly is a mobile commerce application dedicated to discovering and selecting wines. Its aim is to accompany users throughout their journey, from searching for a wine to purchasing it, while providing them with recommendations tailored to their tastes and needs. The application is aimed at both novice consumers, who are looking for simple and accessible advice, and more experienced enthusiasts who want to discover new wines or independent producers.

The app is based on an intuitive user experience, organised around several main screens such as the home page, shop, monthly recommendations, basket and user profile. This structure allows for smooth and logical navigation, facilitating quick access to essential features. Users can explore the wine catalogue, apply personalised filters and obtain suggestions based on their preferences or food and wine pairings.

Sommly also incorporates a mobile commerce dimension by allowing users to add products to their basket and manage their purchases directly from the application. Each wine is associated with a producer, which enhances transparency and promotes a more direct relationship between consumers and sellers.

In addition to its m-commerce features, the app offers integration with connected devices as part of the Internet of Things (IoT). This feature allows users to monitor wine storage conditions, such as temperature and humidity, in real time to ensure optimal quality after purchase. The data collected is securely synchronised and accessible only to authorised users.

Overall, Sommly aims to offer a comprehensive and consistent experience, combining ease of use, personalisation and technological innovation, while adhering to best practices in mobile application design.

IV. Application Architecture and User Flow

The Sommly application architecture is based on a client–server structure adapted to modern mobile applications. The mobile interface constitutes the front-end layer, while the back-end services handle data management, user authentication and real-time information synchronisation. This architecture ensures both the responsiveness of the application and the consistency of the data displayed.

The back-end is based on Firebase, which centralises data related to users, products and interactions within the application. Firebase is also used for authentication management, ensuring secure access to personalised features and sensitive data, particularly that from IoT devices. Real-time synchronisation allows users to immediately view updates, such as changes in wine storage conditions.

Our Firebase instance is organised around three main collections:

- orders: this collection stores all orders placed by users. Each document contains a createdAt timestamp, the order status (paid, pending, etc.), the order items in table form, and information on quantities, prices and selected products. This structure allows us to track purchase history and manage order flow in real time.
- users: this collection contains users' personal information, such as surname, first name, address, city, age, email address, and creation and update timestamps. It allows us to associate this information with users' orders, ensuring a personalised experience.
- wines: this collection contains detailed data on the wines available in the application, including name, colour, description, image, country, region, year, food and wine pairings, price and rating. This information is used to populate the catalogue screens, search filters, recommendations and Sommelier section.

The user journey has been designed to offer simple and intuitive navigation, while limiting the number of steps required to complete the main actions. The user initially accesses the home screen, which displays wine categories, personalised suggestions and the main features of the application. From this screen, they can go to the shop to explore the catalogue or to the recommendation of the month section to see the wines of the month.

When browsing the shop, users can apply different filters to refine their search. Once a wine has been selected, detailed product information is displayed, with the option to add it to the basket. Moving to the basket allows users to check their selection before finalising the purchase, ensuring a smooth and consistent shopping experience.

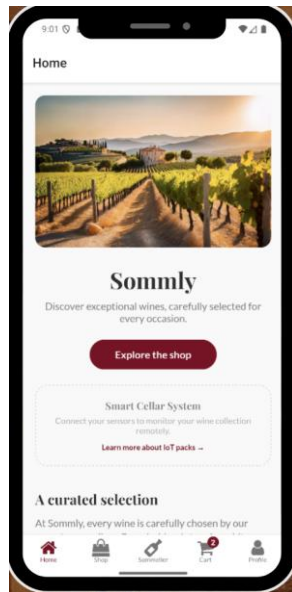
The user profile is a central point in the journey, providing access to personal information, order history and data linked to IoT sensors. This organisation allows advanced features to be grouped together while maintaining a clear separation between the different stages of the journey. Overall, Sommly's architecture and user flow have been designed to ensure a smooth and easy-to-use user experience.

V. Main Screens and Functionality

The Sommly application is structured around several main screens, each with specific features designed to provide a smooth, intuitive and personalised user experience. These screens are directly fed by data stored in Firebase, enabling dynamic, real-time display.

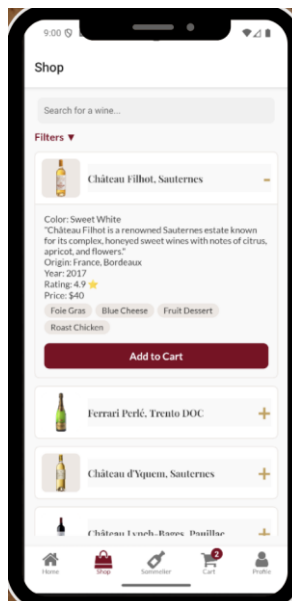
Home Screen

The home screen is the main entry point to the application. It presents users with personalised recommendations based on their preferences and order history, as well as featured wine categories. The suggestions and products displayed are retrieved from the Firebase wines collection, ensuring that the information is always up to date and that images, descriptions and food and wine pairings are displayed correctly.



Shop Screen

The shop allows users to browse the entire wine catalogue. Using advanced filters, users can select wines by type, colour, price, year or food pairings, using data from the wine collection. The results are displayed in a list or grid format, with the image, name, price and rating of each wine. Users can click on a product to access its detailed description.



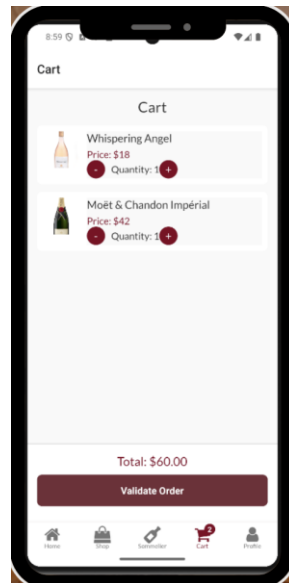
Recommandation Screen

The sommelier section provides personalised recommendations based on user preferences and food and wine pairings available in the wine collection. This feature aims to replicate the experience of consulting a professional sommelier, while remaining simple and accessible. In particular, it allows novice users to receive suggestions tailored to their tastes or the occasion for which they wish to purchase wine.



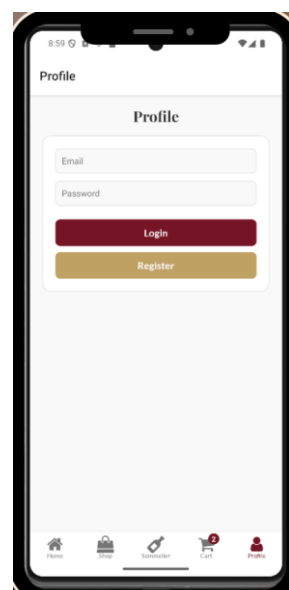
Cart Screen

The cart screen displays the wines selected by the user before finalising the purchase. Each item is populated from the Firebase orders collection and contains information such as the image, name, price, quantity, and order status (paid or pending). The user can change quantities, remove products, or proceed to checkout. Integration with Firebase allows the basket to be updated in real time and ensures synchronisation between different devices.



Profile Screen

The profile screen centralises the user's personal information, such as name, address, city, age and email address, from the useraddress collection. It also provides access to order history and IoT data, if available, allowing users to track the storage conditions of purchased wines. This screen offers a comprehensive and secure view of personal information, while allowing users to update it.



VI. Design Guidelines and UX Principles

The design of the project's user interface was based on a set of design and user experience principles aimed at ensuring intuitive, efficient, and consistent use. One of the main objectives was to propose a clear interface, limiting the cognitive load of the user, notably through a visual prioritization of information and a logical organization of elements. Simplicity has been favored to allow quick handling, even for non-expert users.

Graphic and functional coherence also constituted a central axis of the design. The interactive components (buttons, menus, forms) follow common conventions to ensure immediate understanding of their operation. Furthermore, particular attention has been paid to readability through the choice of colors, contrasts, and typographies, to improve visual comfort and accessibility.

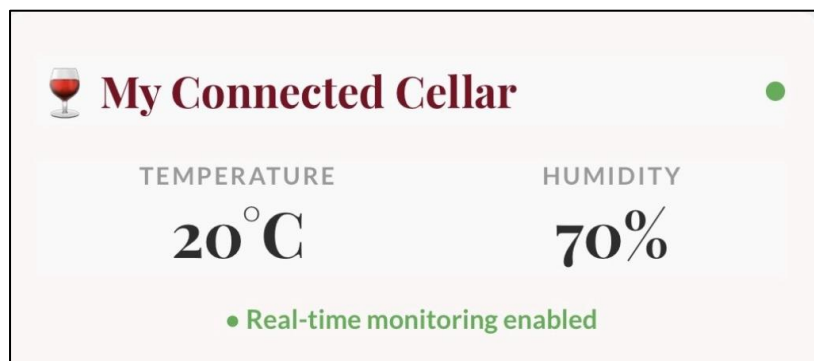
Color palette:



VII. IoT

The Sommly app incorporates Internet of Things (IoT) technology to guarantee wine quality after purchase and improve the user experience. This feature allows users to monitor key wine storage parameters, such as temperature and humidity, in real time. IoT sensors transmit this information to the back end via Firebase, ensuring instant synchronisation with the mobile application and allowing users to view the status of their wines directly from the Profile screen.

Access to this data is secure and restricted to authenticated users, ensuring the confidentiality of sensitive information and personalisation of the experience. With this integration, users can proactively monitor storage conditions and take corrective action if necessary, ensuring that their wines retain their quality and organoleptic characteristics.



This feature also adds value by combining wine discovery and purchase with the ability to monitor its preservation. By tracking environmental conditions in real time, users enjoy a comprehensive and reliable experience, where every step, from selection to preservation, is supported by connected and secure technology.

VIII. Technologie Use

For the development of the mobile application Sommly, several tools and technologies were used to facilitate teamwork, the development of the application and its proper functioning.

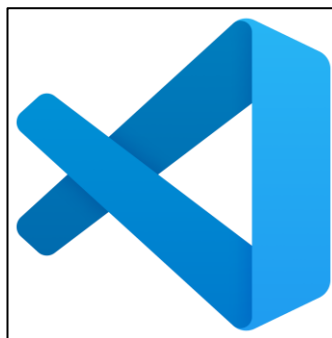
The project was tracked using GitHub, which served to share code and track work progress throughout the project. This tool allowed each team member to access the code, modify certain parts, and keep track of the changes made.



The communication between team members was mainly via WhatsApp, thanks to a group dedicated to the project. This allows for quick exchanges, questions, and easier organization.



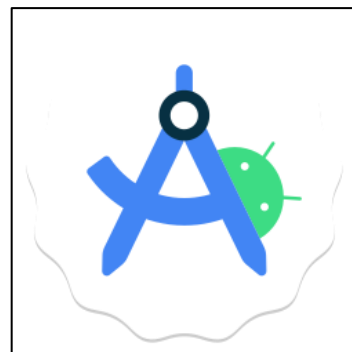
The development of the application was carried out with Visual Studio Code (VS Code). This tool was used to write the project code, because it is easy to use and suitable for mobile application development.



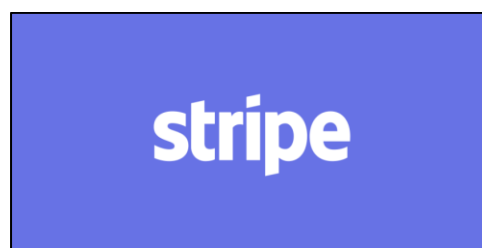
Firebase was used to manage the application's databases. This technology allows storing the data necessary for the operation of Sommly, such as user information and product data, in a reliable and centralized manner.



The application tests were carried out using Expo Go and Android Studio. Expo Go allowed for quick testing of the application on a mobile phone, while Android Studio was used to test the application using an Android emulator.



For the payment part of the application, the Stripe solution has been integrated. It allows you to manage online payments in a secure and reliable manner. It also allows for tracking all orders placed.



IX. Future Improvements

Several developments could be considered to improve the Sommly application and offer an even more complete experience to users. First, adding taste filters would make it easier for customers to find wines that match their preferences. For example, they could select categories such as fruity, dry, light, or more full-bodied wines, which would make the search faster and more intuitive.

The application could also offer wine discovery sessions, led by professional sommeliers. These sessions would give users the opportunity to learn more about the world of wine, grape varieties, tasting methods, and food pairings. In the same logic, Sommly could organize visits to partner wineries, allowing users to discover the origin of the wines offered in the application and to meet the producers.

Moreover, the creation of themed wine packs would be an interesting improvement. These packs could group several bottles around a specific idea, such as the discovery of red and white wines, wines from the same region, or the same grape variety. It would also be possible to offer selections of small portions of exceptional wines, often very expensive per bottle, to make these wines more accessible and allow users to taste them without having to buy a whole bottle.

Finally, Sommly could go further by offering wine creation workshops. During these workshops, participants could discover the different stages of wine making, create their own blend, and taste the wine they have made. This type of activity would strengthen the interactive and experiential aspect of the application, while retaining users around an original and educational approach to wine.

X. Stakeholder Analysis

To ensure the success of Sommly, it was essential to identify and analyze all parties involved or affected by the application. Understanding their needs and influence allowed us to prioritize features and design a truly user-centric experience.

Identification of Key Stakeholders and Strategy

Stakeholder	Description	Influence decision making	Strategy
Primary Users	These include wine enthusiasts, occasional consumers and businesses. These are the heart of the project. They seek a seamless shopping experience, personalized recommendations, and a reliable way to monitor their wine collection via IoT.	High	Manage Closely: Their satisfaction directly impacts the app's success.
Development Team	Our responsibility was to translate user needs into a functional mobile solution, ensuring technical stability and security.	High	Keep Satisfied: Ensuring technical excellence and teamwork.
Wine Producers	They act as the suppliers. Their interest lies in a platform that showcases their craftsmanship and provides a direct sales channel to a diversified audience.	Low	Keep Informed: Providing updates on catalog management and sales tools.